Problem Set 20

Volumes of Revolution I: Disks and Washers

October 26, 2026

Problem 1

(b)

$$V = \pi \int_{r-h}^{r} r^2 - x^2 \, dx = \pi(r^2 x - \frac{x^3}{3})\Big|_{r-h}^{r}$$
$$= \pi(r^3 - \frac{r^3}{3} - r^3 + r^2 h + \frac{r^3}{3} - r^2 h + r h^2 - \frac{h^3}{3})$$
$$= \boxed{\pi h^2 (r - \frac{h}{3})}$$

Problem 2

(b)

$$A = \pi k x$$
$$V = \pi \int_0^a kx \, dx = \pi(\frac{1}{2} k x^2)\Big|_0^a = \boxed{\frac{\pi}{2} k a^2}$$

Problem 4

(a)

$$V = \pi \int_1^2 (2 - \frac{x}{2})^2 \, dx = \pi \int_1^2 \frac{x^2}{4} - 2x + 4 \, dx$$
$$= \pi(\frac{x^3}{12} - x^2 + 4x)\Big|_1^2 = \pi(\frac{7}{12} - 3 + 4) = \boxed{\frac{19\pi}{12}}$$

(b)

$$V = \pi \int_1^2 \frac{1}{x^2} \, dx = \pi(-\frac{1}{x})\Big|_1^2 = \boxed{\frac{\pi}{2}}$$

(c)

$$V = \pi \int_0^9 4y \, dy = 2\pi(y^2)\Big|_0^9 = \boxed{162\pi}$$

(d)

$$V = \pi \int_0^1 x^2 - x^6 \, dx = \pi(\frac{x^3}{3} - \frac{x^7}{7})\Big|_0^1 = \boxed{\frac{4\pi}{21}}$$

(f)

$$V = \pi \int_0^4 (2 - \frac{y}{2})^4 \, dy = \begin{bmatrix} u = 2 - \frac{y}{2} \\ -2du = dy \end{bmatrix} \pi \int_0^2 2u^4 \, dy = 2\pi(\frac{u^5}{5})\Big|_0^2 = \boxed{\frac{64\pi}{5}}$$

Problem 6

The true geometric area is expressed as $A = \pi(R^2 - r^2)$.

$$V = \pi \int_0^1 (x^2 - x^4) \, dx = \pi(\frac{x^3}{3} - \frac{x^5}{5})\Big|_0^1 = \boxed{\frac{2\pi}{15}}$$

$$R^2 - 2Rr + r^2 = R^2 - r^2$$

$$2r(R - r) = 0 \implies \boxed{r = 0 \text{ or } R}$$

Problem 7

(a)

$$V = \pi \int_0^h y \, dy = \boxed{\frac{\pi h^2}{2}}$$

(b)

$$V = \frac{\pi}{2} h^2 \implies dV = \pi h \, dh$$

$$\frac{dh}{dt} = \frac{dV}{dt} \cdot \frac{dh}{dV} = \frac{1}{4\pi} \cdot 2 = \boxed{\frac{1}{2\pi} \text{ cm/s}}$$

(c)

$$V(10) = 50\pi$$

$$25\pi = \frac{\pi}{2} h^2$$

$$h = \boxed{5\sqrt{2} \text{ cm}}$$

The bowl gets wider as the height increases.

Problem 8

(a)

$$(x - R)^2 + y^2 = a^2$$

$$x = \boxed{R \pm \sqrt{a^2 - y^2}}$$

(b)

$$V = \pi \int_{-a}^{a} (R + \sqrt{a^2 - y^2})^2 - (R - \sqrt{a^2 - y^2})^2 \, dy$$

$$= \pi \int_{-a}^{a} 4R\sqrt{a^2 - y^2} \, dy = \pi \cdot 4R \cdot \frac{1}{2}\pi a^2$$

$$= \boxed{2\pi^2 R a^2}$$

(c)

$$2\pi^2 Ra^2 = (2\pi R)(\pi a^2)$$

πa^2: area of the rotated circle.
$2\pi R$: the distance traveled by the center of the circle.

Problem 9

(a) Upper bound: $y = \sqrt{r^2 - x^2}$.

Lower bound: $y = \sqrt{r^2 - \frac{h^2}{4}}$.

(b)

$$\begin{aligned} V &= \pi \int_{-\frac{h}{2}}^{\frac{h}{2}} r^2 - x^2 - r^2 + \frac{h^2}{4}\, dx \\ &= \pi(-\frac{x^3}{3} + \frac{h^2}{4}x)\Big|_{-\frac{h}{2}}^{\frac{h}{2}} \\ &= 2\pi \cdot \frac{2h^3}{24} = \boxed{\frac{\pi h^3}{6}} \end{aligned}$$

(c) The volumes are the same. For the Earth, the radius of the hole is also extremely long, rendering the actual thickness extremely small, which cancels out and leaves an equal volume.

(d)

$$\begin{aligned} V &= 2\pi h^2(r - \frac{h}{3}) \\ &= 2\pi \cdot \frac{1}{4} \cdot \frac{5}{6} \\ &= \boxed{\frac{5\pi}{12}} \end{aligned}$$